A prediction should carry its uncertainty

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Bound / ES: límite / PT: limite

An allowed range, not a standard deviation.

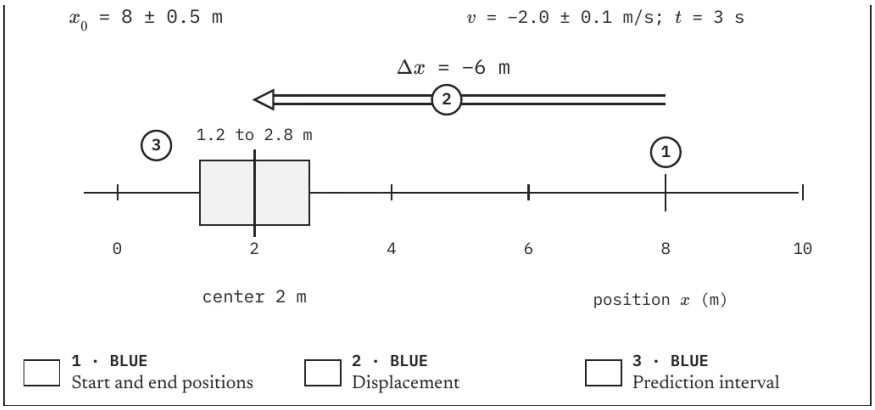
Also on your note page: Initial position, Half-width, Assumption.

B READ THE RELATIONSHIP

Final position = starting position + velocity × time

As you read, find: center · lower end · upper end.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart starts at 8 ± 0.5 m and moves at −2.0 ± 0.1 m/s for an exact 3 s . Find its predicted position interval.

Given x 0 = 8 ± 0.5 m v = −2.0 ± 0.1 m/s t = 3 s ; steady v

Rule Steady v : x f = x 0 + vt . Add input bounds.

Work x f = 8 − 2(3) δ x = .5 + 3(.1) = 0.8 m

Answer 1.2 to 2.8 m 2.0 ± 0.8 m ; full width 1.6 m .

E SAVE FOR THE FINAL MEETING

XJ's position and velocity readings have limits. A forecast should show both the central result and the interval allowed by those limits; the interval does not give an impact probability.

Read the physics: Bound

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Start from the actual starting position. At steady velocity, multiply velocity by time to get the displacement. Add that signed change to the starting position.

02 HOW IT WORKS

Measurements have limits. Mark the central forecast, then the smallest and largest allowed positions. With exact time, allow for the starting-position error and the velocity error multiplied by time.

03 WHAT TO CHECK

Use compatible units before multiplying. A range gives limits; it does not tell you the chance of landing at one particular point.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart starts at 8 ± 0.5 m and moves at -2.0 ± 0.1 m/s for an exact 3 s . Find its predicted position interval. 1.2 to 2.8 m 2.0 ± 0.8 m ; full width 1.6 m .

Model check: Final position = starting position + velocity × time.

05 CONNECT TO XJ

XJ's position and velocity readings have limits. A forecast should show both the central result and the interval allowed by those limits; the interval does not give an impact probability.

SAY IT BEFORE YOU WRITE

What can make the range wider? My center is ___. The allowed positions run from ___ to ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Initial position

ES: posición inicial / PT: posição inicial

Meaning in my words	How can you tell?
How would you explain initial position to a friend? In my words, this means ___.	At 0 s the cart is beside the +5 m mark. Where does its position graph begin? I can tell because ___.
Example or sketch	**Non-example and why**
The cart starts at +5 m and moves 2 m left. What two positions would you mark? My example shows ___.	"The clock starts at zero, so the cart starts at zero." What do the two zeros mean? I would change ___ because ___.

Bound

ES: límite / PT: limite

Meaning in my words	How can you tell?
How would you explain bound to a friend? In my words, this means ___.	For 10 ± 2 km, what is the farthest allowed position on either side? I can tell because ___.
Example or sketch	**Non-example and why**
Mark the bounds for a forecast of 15 ± 3 km. My example shows ___.	A student labels 2 km as the upper endpoint of 10 ± 2 km. What does the 2 mean? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Half-width

ES: semiancho / PT: meia largura

Meaning in my words	How can you tell?
How would you explain half-width to a friend? In my words, this means ___.	A model allows 8 to 12 km. How far is either endpoint from 10 km? I can tell because ___.
Example or sketch	**Non-example and why**
How would you write a range centered on 20 km with half-width 3 km? My example shows ___.	Someone calls 6 km the half-width of 17 to 23 km. What did they measure? I would change ___ because ___.

Assumption

ES: supuesto / PT: hipótese

Meaning in my words	How can you tell?
How would you explain assumption to a friend? In my words, this means ___.	What must be true to predict another 2 m each second from a cart's earlier motion? I can tell because ___.
Example or sketch	**Non-example and why**
Write an assumption beside a straight-line forecast for XJ. My example shows ___.	Is tomorrow's camera reading an assumption once we measure it? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a center tick and both endpoints. Show how starting-position and velocity bounds contribute to the final half-width.

Labels you can use: center · lower end · upper end.

Choose zero, direction, and scale.



Tell a partner

My center is ___. The allowed positions run from ___ to ___.

On assignment: Bound

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two XJ teams predict a center of 20 km. Team A gives ±1 km; team B gives ±4 km. A telescope has one narrow field.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Survey engineer**

Draw both possible bands on one axis. Recommend which report requires a wider search; give both sets of endpoints.

☐ **Weather-desk presenter**

Produce a forecast graphic with two bands. Explain to a viewer which search is wider and what the graphic cannot tell us about impact.

Word bank: center · lower end · upper end.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Distinguish an input bound from a change in the assumed motion.